PROCESSUS COMPLET TECHNIQUE - MINI TASK MANAGER API

1. Preparation Environnement

- Installer Java 17
- Installer Maven
- Installer MySQL
- Installer Docker
- Installer Nexus local
- Installer Ansible sur WSL
- Preparer VM Ubuntu Server

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2. Creation task-core (Maven)

Creer projet Maven avec:

```
GroupId: sn.isi.l3gl.core
ArtifactId: task-core
Version: 0.0.1-SNAPSHOT
```

Structure principale:

```
Task.java
Status.java
TaskRepository.java
TaskService.java
```

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3. Versioning Maven + Nexus

Version 0.0.1-SNAPSHOT -> createTask()

Version 0.1.0-SNAPSHOT -> listTasks()

Version 0.2.0-SNAPSHOT -> updateStatus()

Version 0.3.0-SNAPSHOT -> countCompletedTasks()

Publication SNAPSHOT:

```
mvn clean deploy
```

Publication release 0.3.0:

```
<version>0.3.0</version>
mvn clean deploy
```

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4. Git Workflow task-core

```
git checkout -b develop
git push -u origin develop
```

Feature create-task:

```
git checkout -b feature/create-task
git commit -m "feat: add createTask"
git push origin feature/create-task
```

Merge via Pull Request vers develop.

Repetition pour les autres features.

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5. Creation task-api

```
GroupId: sn.isi.l3gl.api
ArtifactId: task-api
```

Ajouter dependance dans pom.xml:

```
<dependency>
  <groupId>sn.isi.l3gl.core</groupId>
  <artifactId>task-core</artifactId>
  <version>0.3.0</version>
</dependency>
```

Configurer port 8085 dans application.properties:

```
server.port=8085
```

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6. Endpoints REST

```
POST   /api/tasks
GET    /api/tasks
PUT    /api/tasks/{id}/status
GET    /api/tasks/done/count
```

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7. Dockerisation

Dockerfile:

```
FROM openjdk:17-jdk-slim
WORKDIR /app
COPY target/task-api-0.0.1-SNAPSHOT.jar app.jar
EXPOSE 8085
ENTRYPOINT ["java","-jar","app.jar"]
```

Commandes:

```
mvn clean package
docker build -t dockerhub_username/task-api:latest .
docker run -p 8085:8085 dockerhub_username/task-api:latest
docker push dockerhub_username/task-api:latest
```

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8. Ansible Deploiement

inventory.ini:

```
[servers]
192.168.1.10 ansible_user=ubuntu
```

playbook.yml (structure simplifiee):

```
- hosts: servers
  become: yes
  tasks:
    - name: Update apt
      apt:
        update_cache: yes

    - name: Install docker
```

```
    apt:
      name: docker.io
      state: present

  - name: Start docker
    service:
      name: docker
      state: started

  - name: Pull image
    command: docker pull dockerhub_username/task-api:latest

  - name: Run container
    command: docker run -d -p 8085:8085 dockerhub_username/task-api:latest
```

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9. Verification Finale

```
docker ps
curl localhost:8085/api/tasks
```

Verifier via navigateur: http://IP_VM:8085/api/tasks
